**MID SEMESTER EXAMINATION MARCH 2025**

*Name of the Program*: B. TECH — *Semester: IV*

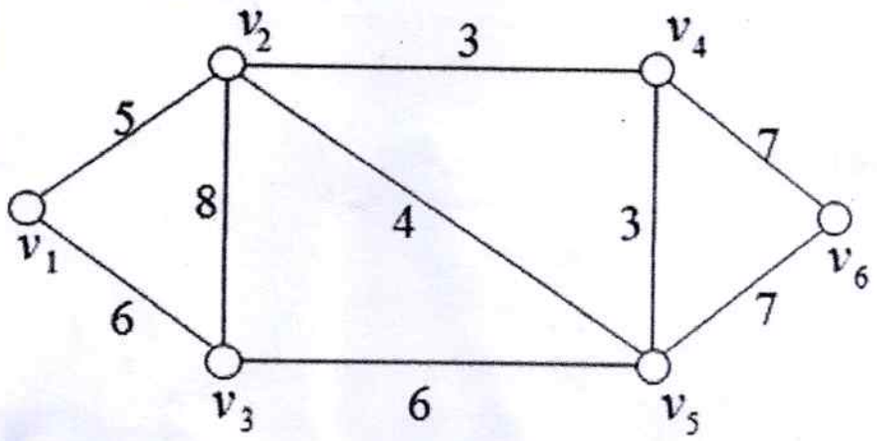
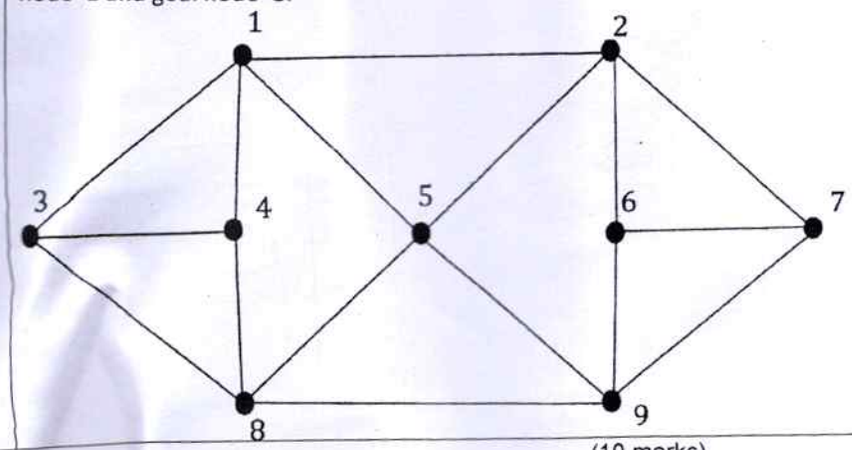
*Name of the Course*: Fundamentals of Statistics & AI — *Course Code:* **TCS 421**

Time: 90 minutes — Maximum Marks: 50

**Note:**

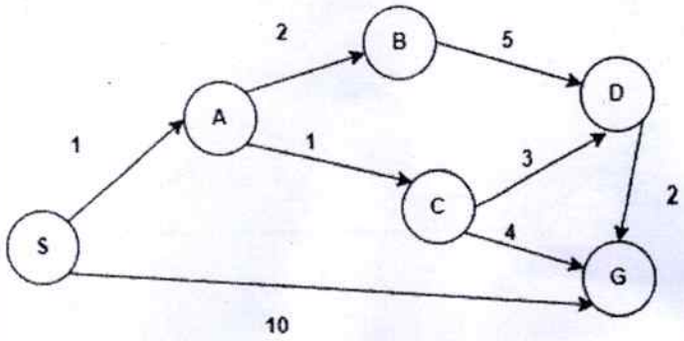
(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

| | | |
|---|---|---|
| Q1 | (10 marks) | |
| a) | Provide a detailed explanation of Data Science, including its core components, methodologies, and tools. Discuss how Data Science is applied across different domains such as healthcare, finance, marketing, manufacturing, and transportation. | |
| | **OR** | |
| b) | Explain BFS method and find the path to reach the goal by applying BFS. Consider start node = $V_1$ and goal node= $V_6$   | CO1,CO5 |
| Q2 | (10 marks) | |
| a) | Define Artificial Intelligence and Explain how it is different from Human Intelligence. | |
| | **OR** | |
| b) | Discuss DFS method and apply the same for the given graph. Consider start node=1 and goal node=8.   | CO1,CO3 |
| Q3 | (10 marks) | CO2,CO5 |

| | | |
|---|---|---|
| a) | XYZ Corp, a leading retail company, is considering implementing AI-driven solutions to enhance customer experience, optimize supply chain operations, and improve sales forecasting. However, the management team is concerned about ethical implications, data privacy, and workforce adaptation. As an AI consultant, how would you guide XYZ Corp in integrating AI effectively? | |
| | **OR** | |
| b) | Write the difference between Informed and Uninformed Searching Algorithms. | |
| Q4 | (10 marks) | |
| a) | Elaborate depth limited search with suitable example. | **CO4,CO5** |
| | **OR** | |
| b) | Why AI is in demand? List the important factors contributing to growth of AI in the real world. | |
| Q5 | (10 marks) | |
| a) | Discuss A* search algorithm and apply the same for the given graph to search the goal node. Consider Start Node=S and Goal Node=G. | **CO1,CO5** |

| S | A | B | C | D | G |
|---|---|---|---|---|---|
| 5 | 3 | 4 | 2 | 6 | 0 |




| | | |
|---|---|---|
| | **OR** | |
| b) | Model Data Science Process using a real time example of your own choice domain. | |